

Modern Business Cycle Theory

Measurement, expectations and a consumption–saving model

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SESSION OBJECTIVE

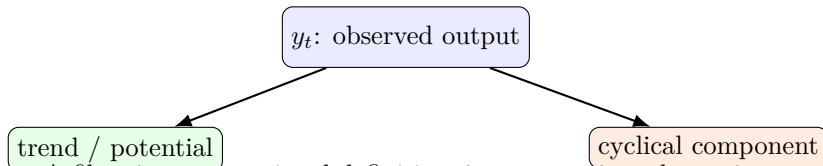
- ▶ Move from observed macroeconomic series to a disciplined notion of a business cycle.
- ▶ Use a simple intertemporal model to connect shocks, expectations and aggregate fluctuations.
- ▶ Prepare the approximation tools used throughout the course.

Handout: Session 1 **Code:** `code/session1/`

PLAN

- 1 What is a cycle?
- 2 A dynamic benchmark
- 3 From nonlinear model to usable system
- 4 Takeaways

OBSERVED OUTPUT COMBINES SEVERAL OBJECTS



A filter is an *operational definition*: its assumptions determine what counts as trend, cycle and measurement noise.

THREE QUESTIONS BEFORE FILTERING

- ① What frequency range corresponds to the economic question?
- ② Is the trend deterministic, stochastic or model based?
- ③ What happens at the sample endpoints and after revisions?

Course principle

Filtering is not neutral preprocessing. It is part of the empirical model and must be documented.

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A CONSUMPTION–SAVING PROBLEM

Household choice:

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E}_0 \sum_{t \geq 0} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = y_t + (1 + r_t)a_t.$$

- ▶ β : patience
- ▶ a_t : assets carried into t
- ▶ r_t : intertemporal price
- ▶ y_t : stochastic income

Economic force: households smooth consumption across states and dates.

EULER EQUATION: THE KEY RESTRICTION

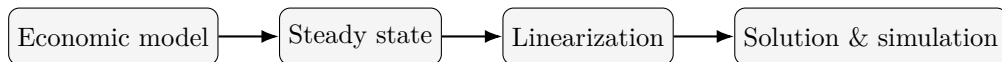
$$u'(c_t) = \beta \mathbb{E}_t [(1 + r_{t+1})u'(c_{t+1})].$$

- ▶ Current consumption trades off against expected marginal utility tomorrow.
- ▶ Expectations are endogenous: beliefs concern future equilibrium objects.
- ▶ Estimation later asks which parameters make these restrictions fit the data.

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A REPRODUCIBLE WORKFLOW



- ▶ Keep the nonlinear model and the approximation separate.
- ▶ Check residuals and units before interpreting an impulse response.

FIRST-ORDER APPROXIMATION

For a system $F(x_{t+1}, x_t, x_{t-1}, \varepsilon_t) = 0$, expand around the deterministic steady state:

$$F_x \hat{x}_{t+1} + F_y \hat{x}_t + F_z \hat{x}_{t-1} + F_\varepsilon \varepsilon_t \simeq 0.$$

Why it matters

Linear systems make expectations, filtering and likelihood evaluation computationally tractable.

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TAKEAWAYS AND PREPARATION

- ▶ Business-cycle facts depend on measurement choices.
- ▶ Dynamic macroeconomic models express restrictions through Euler equations and expectations.
- ▶ Linearization supplies the state-space objects used in Sessions 2–4.

Before Session 2: run `toy_model.m` and inspect the model, steady state and simulated paths.